William Park

236-996-9116 | www.williampark.org | linkedin.com/in/parkwilliamy | github.com/parkwilliamy | w5park@uwaterloo.ca

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Honours Electrical Engineering — *Sept 2024 – May 2029*
Cumulative GPA: 91.09%

Relevant Courses

- Digital Circuits
- Electronic Circuits
- Data Structures
- Numerical Methods
- Digital Computers
- Discrete Math
- Linear Algebra
- Linear Circuits

Technical Skills

Languages: SystemVerilog, Verilog, VHDL, Tcl, Bash, Assembly, Python, C/C++
Developer Tools: Git, Vivado, Quartus Prime, IcarusVerilog, Verilator, gem5, Xschem, SPICE, Ngspice, Magic, OpenOCD, GDB, ROS2, Docker, WSL, Linux, NVIDIA JetPack
Libraries: cocotb, CUDA, MediaPipe, OpenCV, pyserial
Protocols: SPI, I2C, CAN, UART, AXI4

Work Experience

University of Waterloo — May 2026 – Present
Undergraduate Research Co-op — *Waterloo, ON*

- Researching hardware prefetchers for irregular memory access patterns in RISC-V vectorized workloads
- Extending **gem5's** simulation infrastructure in **C++** to prototype and evaluate experimental prefetcher designs

UWASIC — Oct 2025 – Present
Digital Team Lead — *Waterloo, ON*

- Leading a team of students to design and fabricate the first programmable ray-tracing ASIC on Tiny Tapeout
- Designed system architecture to enable parallel development, creating functional blocks, behavioral specifications, and interface definitions
- Profiled ray-tracing kernels to measure arithmetic operation frequency, identifying hardware optimization targets

Kiwi Charge — Sept 2025 – Dec 2025
Electrical Engineering Intern — *Toronto, ON*

- Developed a ROS2-based teleoperation pipeline for a four-wheel differential-drive prototype, integrating joystick input and real-time motor command streaming with the **CAN** bus protocol
- Designed an electrical schematic for the prototype, including power distribution, motor driver interfaces, CAN bus communication, and sensor integration, enabling reliable bring-up and debugging
- Implemented Python scripts to configure and tune multiple motor drivers in parallel over CAN, **reducing setup time by 98%** and improving system scalability

Projects

R500 — June 2025 - Present

- Designed and implemented a **RISC-V** processor in **SystemVerilog** on the Xilinx Artix-7 FPGA
- Reached **100% compliance** with the RV32I ISA based on the official RISC-V compliance test suite
- Achieved a throughput of **46.3 million instructions per second** using a **57 MHz** clock, along with a **CPI of 1.23** and a **branch prediction accuracy of 96.5%**
- Optimized performance through pipelining, forwarding, stalling, flushing, and gshare global branch prediction
- Created unit testbenches in SystemVerilog, employing directed tests and constrained random verification
- Utilized **Tcl** scripts to automate simulation, synthesis, and static timing analysis in Vivado
- Implemented precise exception and interrupt handling in machine mode
- Designing L1 instruction and data caches with shared **AXI4** bus